

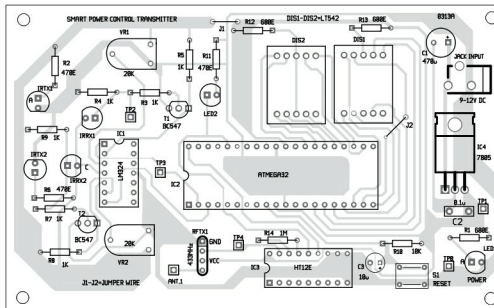


Fig. 5: Component layout for the PCB in Fig. 4

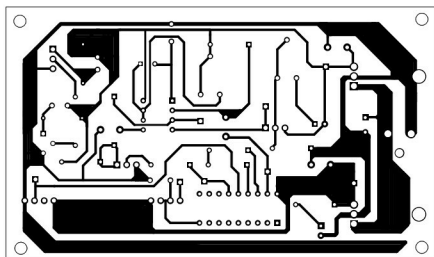


Fig. 6: Actual-size, single-side PCB for the receiver

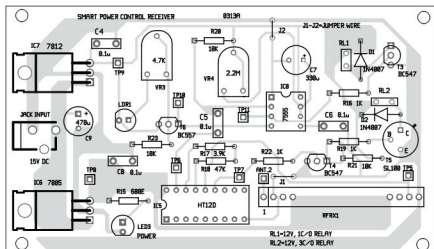


Fig. 7: Component layout for the PCB in Fig. 6

designed specifically for remote controls and wireless security receivers operating at 315/433 MHz. The transmitted signal is received by the receiver antenna (ANT.2) and its data output is fed to pin 14 of HT12D decoder (IC5).

HT12D decoder. It is an 18-pin IC with operating voltage of 2.4V-12V. For proper operation, an encoder-decoder pair with the same number of addresses and data format should be chosen. It receives serial addresses and data from the encoder through RF-RX receiver.

An LED with 1K resistor in series can be connected to pin 17 of HT12D IC, LED will glow whenever a valid signal is received. The data output is available at its pin 13, which is used to switch on/off the MCB power supply line.

There are two power supplies used in the receiver section: 5V and 12V. The 5V supply is used for the receiver including HT12D and RF-RX circuits. The 12V supply is used for relay drivers and relays. Relay RL1 is used to switch on/off the front-door light or corridor light. This light is useful when an employee returns to the office building during night time to search for his missing mobile phone or car key. It remains 'on' for 13 minutes and then goes off. This time duration can be changed using preset VR4.

LDR sensor. Light-dependent resistor LDR1 is used to detect ambient light during day time so that the front-door light does not glow unnecessarily. Sensitivity of the LDR can be set using preset VR3.

RL2 is a 12V, 3C/O relay for three-phase power supply lines. It is the main relay to switch on/off the main MCB. The contact rating of this relay should be as per the rating of the MCB used. If the MCB rating is 32A, relay contact rating should be a minimum of 32A. The current rating of driver